

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #8

Activity Name: Real-Time Clock Interfacing

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Registration No.:	CH.SC.U4CSE24102
Activity Date:	21-08-26
Submission Date:	21-08-26

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	10	

Faculty In-charge

Write an embedded C program for ARM architecture to interface Real-Time Clock. Debug and verify the program using IDE on a development computer system.

```
#include <LPC21xx.h>
```

```
/* LCD Pins */
```

```
#define RS (1 << 16)
```

```
#define EN (1 << 17)
```

```
#define D4 (1 << 18)
```

```
#define D5 (1 << 19)
```

```
#define D6 (1 << 20)
```

```
#define D7 (1 << 21)
```

```
void delay(unsigned int count)
```

```
{  
    unsigned int i, j;  
  
    for(i = 0; i < count; i++)  
    {  
        for(j = 0; j < 1000; j++);  
    }  
}
```

```
/* Send data to LCD */
```

```
void lcd_data(unsigned char data)
```

```
{  
    IOCLR0 = D4 | D5 | D6 | D7;  
  
    if(data & 0x10)  
        IOSET0 = D4;  
  
    if(data & 0x20)  
        IOSET0 = D5;  
  
    if(data & 0x40)  
        IOSET0 = D6;  
  
    if(data & 0x80)  
        IOSET0 = D7;  
  
    IOSET0 = RS;  
  
    IOSET0 = EN;  
    delay(1);  
    IOCLR0 = EN;  
  
    IOCLR0 = D4 | D5 | D6 | D7;
```

```

    if(data & 0x01)
        IOSET0 = D4;

    if(data & 0x02)
        IOSET0 = D5;

    if(data & 0x04)
        IOSET0 = D6;

    if(data & 0x08)
        IOSET0 = D7;

    IOSET0 = EN;
    delay(1);
    IOCLR0 = EN;

    delay(2);
}

/* Send command to LCD */
void lcd_cmd(unsigned char cmd)
{
    IOCLR0 = RS;

    IOCLR0 = D4 | D5 | D6 | D7;

    if(cmd & 0x10)
        IOSET0 = D4;

    if(cmd & 0x20)
        IOSET0 = D5;

    if(cmd & 0x40)
        IOSET0 = D6;

    if(cmd & 0x80)
        IOSET0 = D7;

    IOSET0 = EN;
    delay(1);
    IOCLR0 = EN;

    IOCLR0 = D4 | D5 | D6 | D7;

    if(cmd & 0x01)
        IOSET0 = D4;

```

```

    if(cmd & 0x02)
        IOSET0 = D5;

    if(cmd & 0x04)
        IOSET0 = D6;

    if(cmd & 0x08)
        IOSET0 = D7;

    IOSET0 = EN;
    delay(1);
    IOCLR0 = EN;

    delay(2);
}

void lcd_string(char *str)
{
    while(*str)
    {
        lcd_data(*str);
        str++;
    }
}

void lcd_init(void)
{
    IODIR0 |= RS | EN | D4 | D5 | D6 | D7;

    delay(20);

    lcd_cmd(0x02);

    lcd_cmd(0x28);

    lcd_cmd(0x0C);

    lcd_cmd(0x06);

    lcd_cmd(0x01);

    delay(5);
}

void lcd_number(unsigned int num)

```

```
{  
    lcd_data((num / 10) + '0');  
  
    lcd_data((num % 10) + '0');  
}  
  
/* RTC Initialization */  
void rtc_init(void)  
{  
    CCR = 0x00;  
  
    PREINT = 0x000001C7;  
    PREFRAC = 0x000061C0;  
  
    SEC = 0;  
    MIN = 0;  
    HOUR = 12;  
  
    DOM = 19;  
    MONTH = 8;  
    YEAR = 2026;  
  
    CCR = 0x01;  
}  
  
int main(void)  
{  
    PINSEL0 = 0x00000000;  
    PINSEL1 = 0x00000000;  
  
    lcd_init();  
  
    rtc_init();  
  
    while(1)  
    {  
        /* Display Time */  
  
        lcd_cmd(0x80);  
  
        lcd_string("TIME:");  
  
        lcd_number(HOUR);  
  
        lcd_data(':');
```

```
lcd_number(MIN);
```

```
lcd_data(':');
```

lcd_number(SEC);

```
/* Display Date */
```

```
lcd_cmd(0xC0);
```

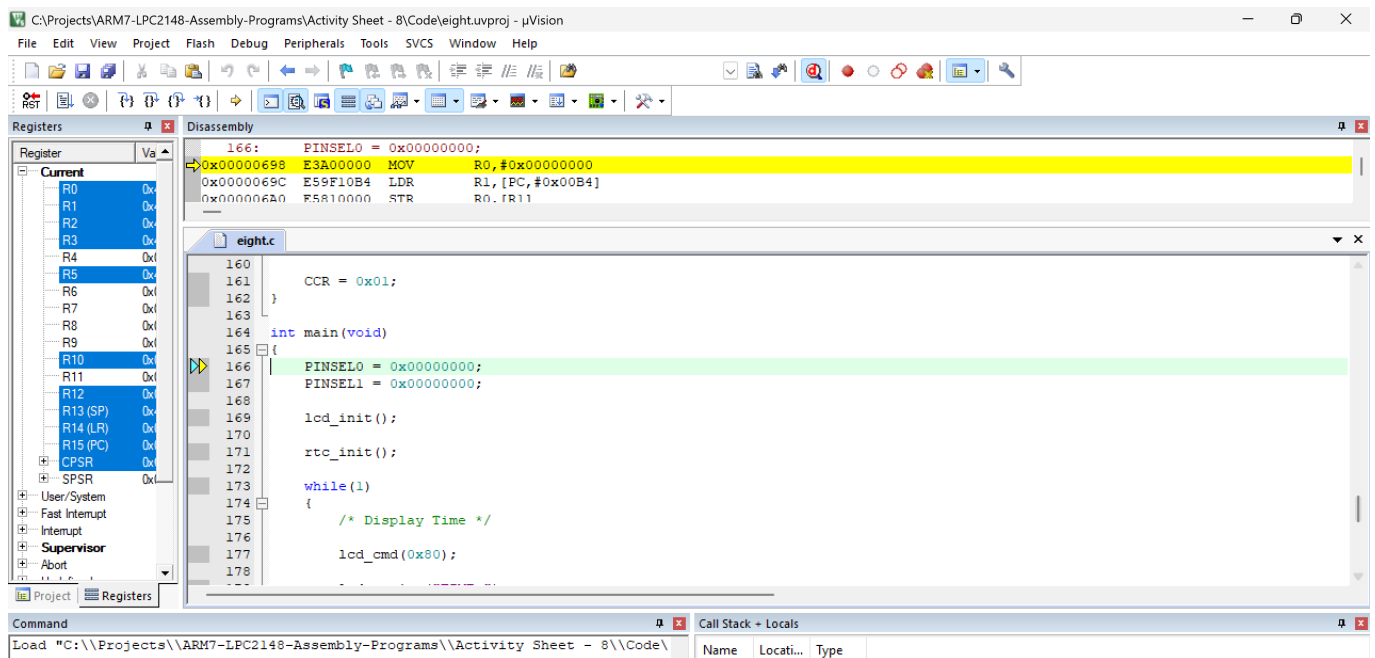
```
lcd_string("DATE:");
```

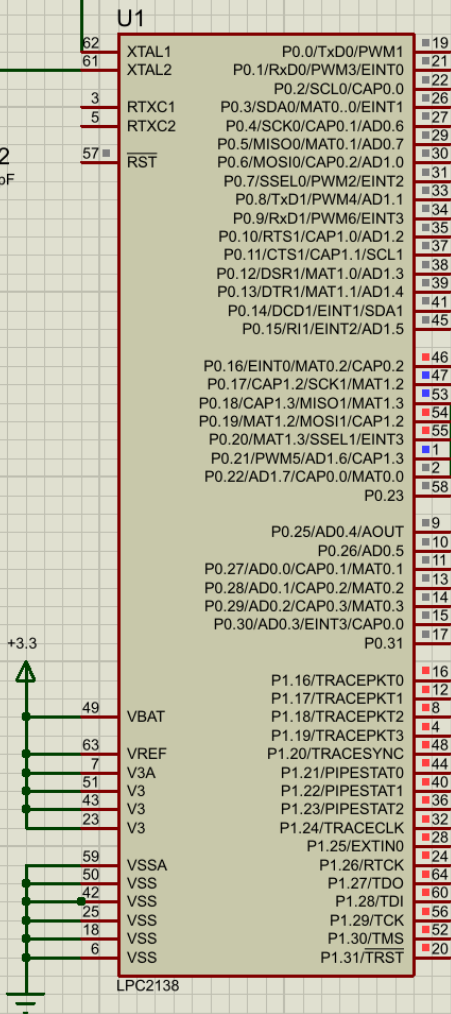
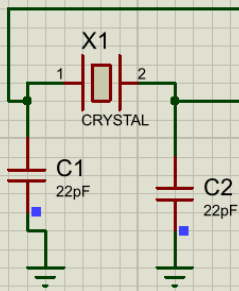
lcd_number(DOM);

```
lcd_data('/');
```

lcd_number(MONTH);**lcd_data('/');****lcd_number(YEAR % 100);****delay(200);**
$$\left. \begin{array}{l} \{ \\ \} \end{array} \right\}$$

OUTPUT:





LCD1
LM016L

